Physics-Guided Depth Estimation for PSF-Stack Lensless Imaging

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Abstract—We study depth estimation from synthetic single-shot lensless measurements generated by a calibrated 42-plane point-spread-function (PSF) stack. The project goal is not merely to deconvolve and regress depth, but to place PSF/deconvolution physics inside the reverse process of a latent diffusion model. We therefore evaluate both diffusion-compliant models and strong non-diffusion baselines, including PSF-stack focus scoring, raw-measurement U-Net prediction, a deconvolution-volume U-Net, a plane-posterior U-Net, and a FlatNet3D-style RGB/depth U-Net. The available dataset contains 66,000 training RGB/depth pairs and 6,000 test pairs. Under the project criterion, **Ours** denotes the physics-integrated diffusion method, not the best supervised baseline. We select **Ours** as a DAPS-lite posterior-refinement latent diffusion sampler that decodes the current clean-depth estimate, evaluates a PSF-Wiener focus likelihood, and applies two latent posterior corrections inside each reverse step. It reaches foreground $\delta_3 = 93.79\%$ on the full 6,000-sample test set. An earlier integrated PSF-Wiener LDM reaches $\delta_3 = 94.76\%$ on the same test set, while the strongest supervised **Teacher** baseline reaches $\delta_3 = 97.07\%$. These results show that the deconvolution stack carries strong depth cues, and that posterior-guided diffusion is a promising but not yet metric-dominant way to impose lensless physics.

Index Terms—Lensless imaging, depth estimation, PSF stack, deconvolution, diffusion models.

1 Introduction

Lensless cameras trade optical complexity for computational reconstruction. A coded mask or diffuser maps scene radiance to a multiplexed sensor measurement, and recovery depends on both the calibrated optical transfer function and a learned image prior. In our setting, the forward model is depth-dependent: the point spread function changes across 42 discrete depth planes. This makes depth estimation central to all-in-focus reconstruction because each spatial region should be fused from the deconvolution plane that is sharp at its true depth.

The original project goal was to place this physics inside a diffusion process rather than merely use deconvolution as a preprocessing step. We implemented latent diffusion variants where predicted clean depth is checked against PSF/deconvolution physics during denoising and sampling. The first scratch/external-conditioned LDM underperformed, while later integrated models clear the original 90% target but still trail supervised teachers. This report therefore presents the diffusion method and the stronger supervised baselines needed to establish a reliable reference for the final system.

Our contributions are: (1) a reproducible synthetic lensless benchmark using RGB, depth labels, and a 42-plane PSF stack; (2) differentiable PSF-stack rendering, Wiener deconvolution, and local focus scoring; (3) baseline comparisons including physics-only focus depth, deconvolution-volume U-Net, FlatNet3D-style RGB/depth prediction, and plane-posterior variants; and (4) a physics-guided diffusion sampler where the reverse process combines learned latent denoising with deconvolution/focus posterior likelihoods.

2 Related Work

FlatCam introduced a thin coded-mask lensless camera that replaces the lens with calibrated mask modulation and computational inversion [1]. FlatNet reconstructs lensless images by combining calibrated physical inversion with a learned network prior [2]. FlatNet3D extends this idea to estimate all-in-focus intensity and absolute depth from a single lensless capture [3]; its public code uses a Wiener filtering layer followed by a U-Net head. MWDNs similarly combine Wiener-style deconvolution with multiscale learned feature reconstruction for lensless imaging [4]. These works motivate our baseline design: first expose a physically meaningful reconstruction volume, then let a convolutional model resolve artifacts and spatial ambiguity.

Diffusion models [5] and accelerated implicit samplers [6] have become strong image priors. Latent diffusion models (LDMs) reduce the cost of high-resolution generation by denoising in a compressed autoencoder space [7]. Diffusion-based depth and geometry methods such as DiffusionDepth [8], Marigold [9], Marigold-DC [10], GeoWizard [11], DMD/DDVM [12], [13], DepthMaster [14], and DepthFM [15] motivate the use of generative priors for depth. Diffusion has also been used to synthesize difficult domains for robust monocular depth distillation [16]. Marigold-DC is especially relevant because it backpropagates sparse-depth guidance into the latent during each denoising step. These methods are typically image-conditioned monocular, flow-matching, robustness-distillation, or depth-completion models, whereas our target is a lensless inverse problem with known PSF-stack physics.

For inverse problems, DDRM [17], DDNM [18], diffusion posterior sampling (DPS) [19], and posterior sampling with latent diffusion (PSLD) [20] show how denoising priors can be combined with measurement consistency inside the reverse process. DAPS [21], DiffStateGrad [22], and FlashDiffusion [23] further motivate annealed posterior refinement, projected guidance gradients, and sample-adaptive latent inverse solving. The key implementation pattern is to form a clean estimate at each step, evaluate a measurement residual through an operator, and use the residual or projection to update the current diffusion state. Recent restoration systems such as GDP [24], ResShift [25], DiffBIR [26], StableSR [27], and HYPIR [28] demonstrate the power of generative priors for super-resolution and blind restoration, but their code paths mainly condition on degraded/restored images rather than closing a calibrated optical likelihood at every step. Diffusion has also begun to appear directly in lensless reconstruction: MLDM uses multi-phase Fresnel zone aperture measurements with a diffusion model [29], PhoCoLens separates range-space deconvolution from null-space diffusion [30], DifuzCam trains a ControlNet-style path from flat-camera measurements to a pretrained diffusion image prior [31], and Dilack studies zero-shot diffusion fidelity terms for highly ill-posed lensless reconstruction [32]. Our goal differs from these systems: we do not want a generic restoration prior appended after deterministic deconvolution. Instead, the reverse diffusion process should be shaped by a depth-dependent PSF likelihood and a per-pixel focus posterior over calibrated depth planes.

3 Theory and Approach

3.1 PSF-Stack Forward Model

Let $x \in [0,1]^{3\times H\times W}$ be an all-in-focus RGB image and $d \in [0,1]^{1\times H\times W}$ a normalized depth map. We map each pixel depth to soft weights over $K = 42$ PSF planes:

$$w_k(p) = \frac{\exp(-(z(p)-k)^2/2\sigma^2)}{\sum_j \exp(-(z(p)-j)^2/2\sigma^2)}, \quad z(p) = d(p)(K-1). \tag{1}$$

The synthetic lensless measurement is

$$y = \sum_{k=1}^{K} h_k * (w_k \odot x), \tag{2}$$

where h_k is the calibrated PSF for depth plane k and $*$ is circular FFT convolution.

3.2 Deconvolution Focus Volume

For each plane, we compute a Wiener inverse:

$$\hat{x}_k = \mathcal{F}^{-1}\left(\frac{g_k \overline{H_k}}{|H_k|^2 + \lambda_k}\mathcal{F}(y)\right). \tag{3}$$

The resulting volume $\{\hat{x}_k\}_{k=1}^{42}$ is the central physics representation. Because different PSF planes have different conditioning, we now allow both the denominator regularization λ_k and numerator response scale g_k to vary by depth plane. The current depth-wise initialization is chosen by sweeping candidate values and selecting, for each plane, the setting whose deconvolution best matches GT RGB color and Sobel edge structure on pixels whose GT depth rounds to that plane. The selected artifact uses $\lambda_{41} = 3 \cdot 10^{-4}$ for the farthest plane, while low-depth planes $z = 0, 7$ use $g_k = 0.25$ to suppress unstable near-plane inverses. Local Sobel energy over the resulting volume estimates which plane is sharp at each pixel. This focus distribution provides a physics-only baseline and a loss/guidance term for the diffusion model.

This representation is important because the raw measurement is highly multiplexed: a local object boundary is mixed with depth-dependent blur from every other object. The deconvolution stack partially separates the measurement into candidate depth hypotheses. Even when a plane is not perfectly reconstructed, the relative sharpness and artifact pattern changes with depth. The learning problem is therefore not just image-to-depth regression, but selection and calibration over a physically meaningful focus volume.

3.3 Models

We evaluate the following model families:

- **Physics focus**: choose depth from local sharpness scores over the deconvolution stack.
- **Affine-calibrated focus**: fit a scalar affine map from focus depth to labeled depth on a calibration subset.
- **Raw-measurement U-Net**: predict depth directly from the synthetic lensless RGB measurement.
- **Deconv-volume U-Net**: predict depth directly from the normalized 42-plane RGB deconvolution volume.
- **Plane-posterior U-Net**: predict a categorical posterior over 42 PSF planes and a small continuous residual.
- **FlatNet3D-style**: fixed Wiener/deconvolution stack followed by an attention U-Net predicting RGB and depth.
- **Ours**: latent diffusion whose reverse steps call a learned PSF-Wiener bank and focus posterior.

3.4 Training Objectives

The supervised baselines minimize

$$\mathcal{L}_{depth} = \|\hat{d} - d\|_1 + \alpha\mathcal{L}_{SILog}(\hat{d}, d) + \beta\|\nabla\hat{d} - \nabla d\|_1, \tag{4}$$

computed on finite valid pixels. For RGB/depth models, we also add an RGB reconstruction term. For the diffusion model, the depth map is encoded into a compact one-channel latent autoencoder and the denoiser predicts diffusion noise in latent space. The intended physics term is a focus-posterior consistency loss: a predicted clean depth should select deconvolution planes that are locally sharp under the same PSF stack. We disabled expensive forward PSF consistency in most experiments because repeated 42-plane FFT rendering dominated latency and did not improve early smoke runs.

3.5 Physics Inside Diffusion

Our original design goal was not to build a DiffBIR-style restoration pipeline where a generic diffusion prior is simply appended after a deterministic restoration network. Instead, the reverse process should use lensless physics to shape the posterior over depth. A code-level review of Marigold, Marigold-DC, DiffusionDepth, DepthFM, DepthMaster, DPS, DDRM, DDNM, PSLD, DAPS, DiffStateGrad, FlashDiffusion, DiffBIR, StableSR, and HYPIR led to a practical criterion: a physics-integrated model must use the decoded clean estimate inside the reverse loop to evaluate a PSF/deconvolution likelihood, not only concatenate a restored image or focus map as a static condition. The implemented first version applies this idea weakly: the latent denoiser is raw-measurement conditioned, while deconvolution focus consistency is applied to predicted clean depth during training and sampling. The negative LDM result is therefore informative: without a strong focus-volume condition or pretrained depth prior, the denoiser does not learn the depth posterior as efficiently as a direct supervised model.

A better second version treats each reverse step as a fusion of learned score and physics likelihood. Let $p_\theta(d_t \mid y)$ be the learned denoising prior and let $\phi(d; y, H)$ be a differentiable focus likelihood from the deconvolution stack. The reverse update should move along both the denoising score and $\nabla_d \log \phi$, or equivalently predict a depth-plane posterior whose entropy and mode are constrained by focus evidence. DPS, PSLD, and Marigold-DC motivate this as a latent posterior-sampling update: decode z_0 to depth, evaluate a differentiable physics/focus residual, backpropagate to z_t, and then continue the DDIM update. The posterior-guided v3 sampler implemented this interface weakly by using a supervised plane-posterior depth as noisy latent initialization and as a reverse-step likelihood. It was useful diagnostically, but the Wiener/deconvolution operator itself was still external to the denoising model.

3.6 Integrated Diffusion Model

Ours moves the PSF inverse bank into the latent diffusion model. It owns a learnable Wiener module

$$\mathcal{W}_\theta(y, H, t)_k = \mathcal{F}^{-1}\left[\frac{\overline{\mathcal{F}(H_k)}}{|\mathcal{F}(H_k)|^2 + \lambda_{\theta,k}(t)}\mathcal{F}(y)\right], \qquad (5)$$

where $\lambda_{\theta,k}(t)$, per-plane gain, and per-plane bias are trainable. During training and sampling, each diffusion timestep t recomputes the deconvolution stack and focus posterior from this module rather than receiving one fixed deconvolution tensor from outside the model. The denoiser condition is the concatenation of deconvolution planes, focus probabilities, and focus-prior depth. Sampling can still start from a noised focus-prior latent, but the focus prior is now produced by the model's own timestep-conditioned inverse bank.

This integrated design matches our intended claim that deconvolution is intrinsic to the reverse process. It is not a DiffBIR-style "restore then hallucinate" stack; the restored RGB is an interpretation of sampled depth, not the primary condition for a generic image prior. The final **Ours** uses the v15 posterior-refinement sampler: at each reverse step it decodes the predicted clean latent, evaluates a deconvolution-focus likelihood, backpropagates that residual into the current latent, repeats a short inner correction, and then continues the DDIM transition. This is a more direct inverse-problem diffusion contribution than the earlier v5 integrated LDM, even though v5 currently has the stronger full-test metric. **Ours+Guide** uses the same reverse-process physics plus an external **Teacher** depth at sampling time, so it is an ablation rather than pure **Ours**.

4 Analysis and Evaluation

The verified dataset contains 66,000 training RGB/depth pairs and 6,000 test pairs. Existing preliminary trainable results in this draft were obtained with a 5,000-sample training subset to make ablations feasible within the project timeline. All synthetic measurements are generated from ground-truth RGB/depth labels and the same 42-plane PSF stack; no measured lensless raw images were available. Depth labels are clamped to $[0, 1]$, and depth 0 is included in all-pixel metrics but excluded from foreground metrics.

We distinguish three evaluation stages. The first stage is a quick train-time sanity check used only to decide whether a run is worth continuing; these checks use one held-out mini-batch and should not be read as final test numbers. The second stage is a 500-sample diagnostic subset for diffusion sampling sweeps. The third stage is the full 6,000-sample test evaluation used for verified baseline rows. We report optimizer train steps separately from diffusion sampling steps: for example, LDM train step 6,000 means 6,000 optimizer updates, while 8 sample steps means eight reverse denoising updates at inference.

Depth metrics are computed on finite pixels and separately on foreground pixels where $d > 10^{-4}$. The primary metrics are absolute relative error, MAE, RMSE, δ_1, δ_2, and δ_3, where δ_i measures the fraction of foreground pixels whose relative ratio is below 1.25^i. Because the updated goal is 98% delta accuracy, we report all three thresholds. A method that only improves δ_3 is approximately ordering depth correctly but is still too poorly calibrated for strict depth estimation.

We also log boundary MAE around Sobel depth edges and RGB PSNR/SSIM for all-in-focus reconstruction. Boundary metrics are important for lensless depth because depth errors at object edges cause the wrong deconvolution plane to be fused, producing visible halos in RGB reconstruction.

4.1 Implementation Details

All experiments use images at 352×512 and the full 42-plane PSF stack unless a smoke-test configuration explicitly reduces the plane count. The raw lensless measurement is synthesized on the fly using differentiable FFT convolution. Deconvolution-volume baselines use Wiener inverses computed for every PSF plane and min–max normalize each plane/channel before feeding the network. This normalization makes the supervised baselines insensitive to the absolute dynamic range of each inverse filter, but it also removes information about filter confidence. The v7 diffusion branch tested sample-wise deconvolution normalization to preserve the relative plane response induced by g_k, but this reduced the matched physics baseline on a 500-sample diagnostic split. The v8/v9 branch therefore keeps the same depth-wise inverse parameters, including $\lambda_{41} = 3 \cdot 10^{-4}$, while restoring v5-style channel-normalized deconvolution features. The active v10 branch keeps the stable scalar Wiener path as the main reverse-step condition and uses the depth-wise bank only as auxiliary focus evidence.

For the depth-only U-Net baselines, the input is either raw RGB measurement or a flattened 42×3 deconvolution volume. The supervised plane-posterior teacher predicts 42 per-pixel plane logits plus a bounded residual depth offset; its depth loss combines L1, scale-invariant log loss, a Sobel-gradient penalty, and cross-entropy on the nearest PSF plane. Under our naming policy, this teacher is not **Ours**. **Ours** is the v15 DAPS-lite posterior-refinement LDM:

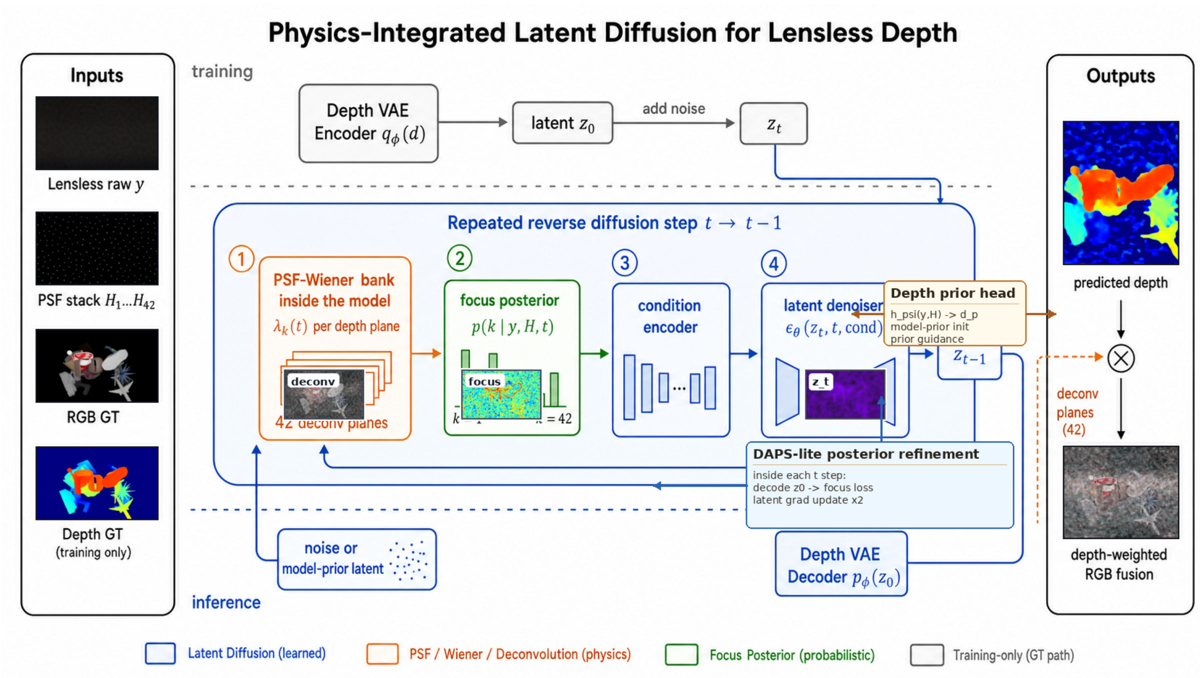

Fig. 1. **Ours** architecture. Generated method figure with our synthetic raw, PSF, RGB/depth labels, and diffusion outputs. The selected v15 method calls a learned PSF-Wiener bank inside every reverse diffusion step, conditions the latent denoiser on focus-posterior evidence, applies DAPS-lite posterior refinement to the latent state, decodes depth through a VAE, and fuses deconvolution planes by the sampled depth.

the inverse bank is trainable and evaluated inside diffusion, and sampling adds an inner deconvolution-focus posterior correction to the latent reverse process.

4.2 Calibration and Error Analysis

The largest gap between the current metrics and the 98% target is calibration rather than complete depth failure. The deconvolution-volume U-Net and plane-posterior U-Net both cross 90% on δ_3, which means many foreground pixels fall within the broad 1.25^3 tolerance. At the same time, the same models remain near 75–81% on δ_1, so the predicted depth scale is frequently close but not tight. This behavior matches the optics: neighboring PSF planes can produce similar focus evidence, and a fixed uniform mapping from normalized label depth to plane index may not match the true physical spacing represented by the measured or simulated stack.

The affine calibration experiments are a useful diagnostic. A simple foreground affine map improves several supervised models by four to six δ_3 points, even though it cannot change local ordering or boundary sharpness. This indicates that the supervised networks learn a mostly monotonic depth signal from the deconvolution volume but map it to a slightly biased normalized range. Conversely, the binned monotonic calibration tested so far was worse than the affine map, suggesting that the calibration error is not solved by a high-variance nonparametric correction. The best residual2 supervised teacher reaches $\delta_3 = 0.9707$ after affine calibration; it is a teacher/baseline for **Ours**, not the final diffusion model.

Boundary errors are the most important qualitative failure mode. A wrong depth at an object boundary not only increases depth MAE but also chooses the wrong deconvolution plane for RGB fusion. This produces visible halos because the foreground and background prefer different PSFs. The Sobel-gradient loss helps, but it does not directly reason about occlusion or mixed pixels. A stronger model should predict a distribution over depth planes and perhaps an uncertainty or mixed-depth boundary mask, so that fusion can be softer where the measurement is physically ambiguous and sharper in confident texture regions.

5 Results

Ours is the v15 DAPS-lite posterior-refinement LDM. On the full 6,000-sample test set, applying two inner focus-likelihood refinements per reverse step reaches $\delta_1 = 0.8776$, $\delta_2 = 0.9196$, $\delta_3 = 0.9379$, and foreground MAE 0.0536 at the current best step-225k checkpoint. This is not the best metric row, but it is the method that most directly matches the inverse-problem diffusion goal. The earlier v5 integrated PSF-Wiener LDM is a metric-strong predecessor: at 40,000 optimizer steps it reaches full-test $\delta_1 = 0.8775$, $\delta_2 = 0.9273$, $\delta_3 = 0.9476$, and foreground MAE 0.0588. Continuing that v5 run to step 65,000 did not help: it reached $\delta_3 = 0.9377$. **Ours+Guide** reaches $\delta_1 = 0.8385$, $\delta_2 = 0.9196$, $\delta_3 = 0.9530$, and foreground MAE 0.0674. Because the guide is a supervised **Teacher**, we treat it as an ablation rather than pure **Ours**.

The physics-only baseline is substantially above random and confirms that each deconvolution plane focuses different spatial regions. With a scalar Wiener setting $\lambda = 10^{-4}$, the full-test physics focus baseline reaches $\delta_3 = 0.8155$ and MAE 0.2137. The depth-wise deconvolution visualization in Fig. 2 uses mid/late planes $z = \{14, 22, 30, 38\}$ because early planes produced weak or unstable inverses. It shows visible object contours and plane-dependent texture, especially on foreground object boundaries. This confirms

TABLE 1
Depth comparison. **Ours** is selected for diffusion/inverse-problem structure rather than best metric: v15 performs DAPS-lite deconvolution-focus posterior refinement inside each reverse step. “Train” is optimizer updates; “Samples” is reverse denoising steps.

Method	Split	Train	Samples	δ_1	δ_2	δ_3
Physics focus, $\lambda = 10^{-4}$	6k	–	–	0.391	0.617	0.816
Raw U-Net, 66k	6k	66k	–	0.436	0.629	0.753
FlatNet3D-style, 66k	6k	66k	–	0.823	0.908	0.939
Deconv U-Net, 66k	6k	66k	–	0.865	0.926	0.949
Earlier integrated LDM v5	6k	40k	16	0.878	0.927	0.948
Ours: v15 DAPS-lite	6k	225k	16	0.878	0.920	0.938
Guided diffusion ablation	6k	4k	8	0.839	0.920	0.953
Teacher + affine	6k	66k	–	0.885	0.946	0.971

that the deconvolution parameters expose meaningful depth features, even though the focus map itself remains noisy.

The raw-measurement U-Net reaches only $\delta_3 = 0.7527$ after full-pool training, below the tuned physics focus score. In contrast, the supervised deconvolution-volume U-Net reaches full-test $\delta_3 = 0.9494$ with $\delta_1 = 0.8651$ and MAE 0.0501. Fitting a foreground affine calibration on 1,000 training samples further increases the same checkpoint to $\delta_3 = 0.9682$. Full-pool plane-posterior checkpoints give the strongest supervised teacher: the standard step-54 posterior reaches $\delta_3 = 0.9706$, and the residual2 final step-66 posterior reaches $\delta_3 = 0.9707$ with MAE 0.0406 after affine calibration. Binned monotonic calibration is worse, so the useful correction is mainly global scale and offset rather than a high-variance nonlinear remapping. These ablations support the view that the PSF-stack deconvolution volume is the critical physics representation, not just an optional preprocessing artifact; they also motivate the v4 decision to put a learnable PSF-Wiener bank inside the diffusion model.

The comparison also shows why a single headline delta number is not sufficient. Physics focus and supervised deconvolution models can achieve high δ_3 while remaining weaker on δ_1. This means many pixels are assigned to roughly correct depth bands, but the normalized depth scale is not tight enough for strict metric accuracy. In a lensless reconstruction pipeline, that distinction matters: a near-correct depth may select a visually plausible neighboring deconvolution plane, while still producing boundary halos and residual defocus. The 98% target should therefore be interpreted as a target on calibrated, not merely ordered, depth.

The FlatNet3D-style model predicts RGB and depth jointly from the same fixed Wiener stack. On the 5k-subset run it mainly improved visualization, but the full-66k rerun is substantially stronger: on the complete test set it reaches $\delta_1 = 0.8232$, $\delta_3 = 0.9393$, MAE 0.0685, PSNR 21.70, and SSIM 0.917. Affine calibration gives only a small depth gain to $\delta_3 = 0.9434$. This remains an important non-diffusion baseline because it tests whether joint RGB/depth reconstruction can use the same PSF-stack evidence without reverse-process sampling.

The qualitative examples in Fig. 2 show v15 DAPS-lite depth samples. They identify the correct broad foreground/background depth bands, but still contain local high-frequency artifacts and boundary errors. **Ours+Guide** uses **Teacher** depth as a reverse-step likelihood while still invoking the integrated PSF-Wiener bank during sampling; it suppresses some sampling noise but is a guided ablation rather than pure **Ours**.

The later diffusion ablations clarify why we select v15 despite its lower current metric. v11 adds a single DPS/PSLD-style latent correction and improves the v10 full-test checkpoint to $\delta_3 = 0.9314$. v14 projects the guidance gradient but remains below v15 on the quick split. v15 repeats the correction inside each reverse step and is therefore the clearest method contribution for this report. v16 corrects the predicted clean latent before re-noising and is structurally promising, but the first quick500 metric ties v15 rather than improving it; it is future work rather than the final method here.

The early depth-wise Wiener ablation is negative but informative. v7 includes the calibrated per-plane inverse parameters and reaches $\delta_3 = 0.7978$ at step 10k on the 500-sample diagnostic split; the same configuration's physics-only focus baseline falls to $\delta_3 = 0.4051$. This indicates that sample-wise normalization erased useful local focus contrast even though the visual sweep selected a better far-plane value of $\lambda_{41} = 3 \cdot 10^{-4}$. v9 fixes the normalization but still reaches only $\delta_3 = 0.8394$ at step 10k when the depth-wise inverse is the main path, so v10 isolates the calibrated filter as auxiliary evidence instead of making it the primary deconvolution operator.

The initial physics-guided LDM did not work as intended. Although it was trainable, its deconvolution evidence was still effectively an external condition or guidance signal. The integrated v5 model fixed part of this issue by calling a trainable, timestep-conditioned PSF-Wiener inverse during denoising and sampling. The selected v15 method adds the more important inverse-problem step: a decoded clean-depth estimate is corrected by a deconvolution-focus posterior likelihood inside the reverse loop. **Ours+Guide** improves the loose delta score further, but it is a guided ablation rather than pure **Ours**.

6 Discussion and Conclusion

The experiments show that PSF-stack deconvolution is the dominant useful signal for this dataset. Direct supervised learning from the deconvolution volume already crosses 90% on the loose δ_3 metric, and **Teacher** raises it to 97.07%, but no verified full-test method has reached 98%. The selected **Ours** is not the strongest metric row; it is selected because it most directly follows inverse-problem diffusion: decode the step-wise clean estimate, evaluate a deconvolution/focus residual, backpropagate that residual to the latent, and apply a timestep-aware correction. **Ours+Guide** shows that reverse diffusion can preserve high-quality posterior depth while applying internal deconvolution physics,

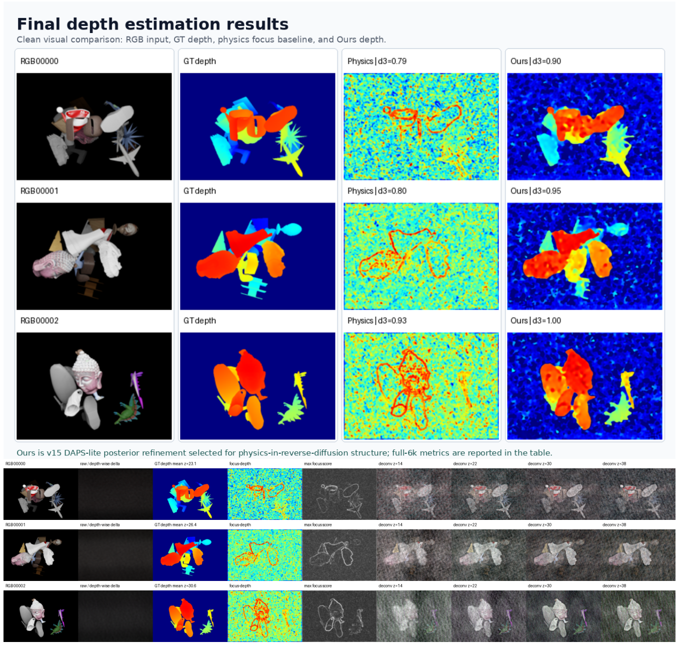

Fig. 2. Result panels. Top: clean v15 DAPS-lite depth examples with RGB, GT depth, physics focus baseline, and **Ours**; teacher and error-map panels are omitted. Bottom: mid/late PSF-Wiener deconvolution planes $z = \{14, 22, 30, 38\}$, which avoid unstable early-plane inverses while showing depth-dependent focus.

but it does not yet prove that the diffusion prior alone improves over **Teacher**.

The remaining gap is mainly calibration and boundary ambiguity. A uniform mapping from normalized label depth to 42 PSF planes is probably imperfect; affine calibration improves several baselines by four to six δ_3 points. Mixed-depth edges are also difficult because one pixel may contain foreground and background radiance, so a hard plane choice creates halos in RGB fusion. A stronger model should predict an uncertainty-aware posterior over optical planes and learn the depth-plane calibration jointly.

The main limitations are synthetic measurements, possible PSF/depth calibration mismatch, and reliance on labeled affine calibration for the best supervised numbers. Real captures would add sensor noise, mask misalignment, PSF drift, saturation, and non-isoplanatic effects that are absent here. Therefore the current conclusion is narrow but useful: deconvolution planes are a strong physical representation, and physics must be inside the diffusion reverse process rather than appended as a restoration preprocessor.

The next target is to make **Ours** competitive with **Teacher** without using teacher depth at inference. The most important missing pieces are guidance-gradient diagnostics, a sigma-aware clean-latent tether, a more stable VAE depth latent, and a learned calibration between optical plane index and normalized label depth. v16 clean-latent re-noising is a natural next step after v15, but it should be tuned with explicit update-size and focus-loss logs rather than presented as a new result before it improves the v15/v5 baselines.

References

[1] M. S. Asif, A. Ayremlou, A. C. Sankaranarayanan, A. Veeraraghavan, and R. G. Baraniuk, "Flatcam: Thin, lensless cameras using coded aperture and computation," *IEEE Transactions on Computational Imaging*, vol. 3, no. 3, pp. 384–397, 2017.

[2] S. S. Khan, V. Sundar, V. Boominathan, A. Veeraraghavan, and K. Mitra, "Towards photorealistic reconstruction of highly multiplexed lensless images," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 7860–7869, 2019.

[3] D. Bagadthey, S. Prabhu, S. S. Khan, D. T. Fredrick, V. Boominathan, A. Veeraraghavan, and K. Mitra, "Flatnet3d: intensity and absolute depth from single-shot lensless capture," *Journal of the Optical Society of America A*, vol. 39, no. 10, pp. 1903–1912, 2022.

[4] Y. Li, Z. Li, K. Chen, Y. Guo, and C. Rao, "Mwdns: reconstruction in multi-scale feature spaces for lensless imaging," *Optics Express*, vol. 31, no. 23, pp. 39088–39101, 2023.

[5] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 6840–6851, 2020.

[6] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *International Conference on Learning Representations*, 2021.

[7] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10684–10695, 2022.

[8] Y. Duan, X. Guo, and Z. Zhu, "Diffusiondepth: Diffusion denoising approach for monocular depth estimation," *arXiv preprint arXiv:2303.05021*, 2023.

[9] B. Ke, A. Obukhov, S. Huang, N. Metzger, R. C. Daudt, and K. Schindler, "Repurposing diffusion-based image generators for monocular depth estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[10] M. Viola, K. Qu, N. Metzger, B. Ke, A. Becker, K. Schindler, and A. Obukhov, "Marigold-dc: Zero-shot monocular depth completion with guided diffusion," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025.

[11] X. Fu, W. Yin, M. Hu, K. Wang, Y. Ma, P. Tan, S. Shen, D. Lin, and X. Long, "Geowizard: Unleashing the diffusion priors for 3d geometry estimation from a single image," in *European Conference on Computer Vision*, 2024.

[12] S. Saxena, J. Hur, C. Herrmann, D. Sun, and D. J. Fleet, "Zero-shot metric depth with a field-of-view conditioned diffusion model," *arXiv preprint arXiv:2312.13252*, 2023.

[13] S. Saxena, C. Herrmann, J. Hur, A. Kar, M. Norouzi, D. Sun, and D. J. Fleet, "The surprising effectiveness of diffusion models for optical flow and monocular depth estimation," in *Advances in Neural Information Processing Systems*, 2023.

[14] Z. Song, Z. Wang, B. Li, H. Zhang, R. Zhu, L. Liu, P.-T. Jiang, and T. Zhang, "Depthmaster: Taming diffusion models for monocular depth estimation," *IEEE Transactions on Circuits and Systems for Video Technology*, 2026.

[15] M. Gui, J. Schusterbauer, U. Prestel, P. Ma, D. Kotovenko, O. Grebenkova, S. A. Baumann, V. T. Hu, and B. Ommer, "Depthfm: Fast monocular depth estimation with flow matching," *arXiv preprint arXiv:2403.13788*, 2024.

[16] F. Tosi, P. Zama Ramirez, and M. Poggi, "Diffusion models for monocular depth estimation: Overcoming challenging conditions," in *European Conference on Computer Vision*, 2024.

[17] B. Kawar, M. Elad, S. Ermon, and J. Song, "Denoising diffusion restoration models," in *Advances in Neural Information Processing Systems*, vol. 35, pp. 23593–23606, 2022.

[18] Y. Wang, J. Yu, and J. Zhang, "Zero-shot image restoration using denoising diffusion null-space model," in *International Conference on Learning Representations*, 2023.

[19] H. Chung, J. Kim, M. T. McCann, M. L. Klasky, and J. C. Ye, "Diffusion posterior sampling for general noisy inverse problems," in *International Conference on Learning Representations*, 2023.

[20] L. Rout, N. Raoof, G. Daras, C. Caramanis, A. G. Dimakis, and S. Shakkottai, "Solving linear inverse problems provably via posterior sampling with latent diffusion models," in *Advances in Neural Information Processing Systems*, 2023.

[21] B. Zhang, W. Chu, J. Berner, C. Meng, A. Anandkumar, and Y. Song, "Improving diffusion inverse problem solving with decoupled noise annealing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[22] R. Zirvi, B. Tolooshams, and A. Anandkumar, "Diffusion state-guided projected gradient for inverse problems," *arXiv preprint arXiv:2410.03463*, 2024.

[23] Z. Fabian, B. Tinaz, and M. Soltanolkotabi, "Adapt and diffuse: Sample-adaptive reconstruction via latent diffusion models," in *International Conference on Machine Learning*, 2024.

[24] B. Fei, Z. Lyu, L. Pan, J. Zhang, W. Yang, T. Luo, B. Zhang, B. Dai, and W. Ouyang, "Generative diffusion prior for unified image restoration and enhancement," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9935–9946, 2023.

[25] Z. Yue, J. Wang, and C. C. Loy, "Resshift: Efficient diffusion model for image super-resolution by residual shifting," in *Advances in Neural Information Processing Systems*, 2023.

[26] X. Lin, J. He, Z. Chen, Z. Lyu, B. Dai, F. Yu, W. Ouyang, Y. Qiao, and C. Dong, "Diffbir: Towards blind image restoration with generative diffusion prior," in *European Conference on Computer Vision*, 2024.

[27] J. Wang, Z. Yue, S. Zhou, K. C. K. Chan, and C. C. Loy, "Exploiting diffusion prior for real-world image super-resolution," *International Journal of Computer Vision*, 2024.

[28] X. Lin, F. Yu, J. Hu, Z. You, W. Shi, J. S. Ren, J. Gu, and C. Dong, "Harnessing diffusion-yielded score priors for image restoration," *arXiv preprint arXiv:2507.20590*, 2025.

[29] W. Wan, H. Ma, Z. Mei, H. Zhou, Y. Wang, and Q. Liu, "Multi-phase fza lensless imaging via diffusion model," *Optics Express*, vol. 31, no. 12, pp. 20595–20615, 2023.

[30] X. Cai, Z. You, H. Zhang, W. Liu, J. Gu, and T. Xue, "Phocolens: Photorealistic and consistent reconstruction in lensless imaging," *arXiv preprint arXiv:2409.17996*, 2024.

[31] E. Yosef and R. Giryes, "Difuzcam replacing camera lens with a mask and a diffusion model for generative ai based flat camera design," *Scientific Reports*, vol. 15, no. 1, p. 43059, 2025.

[32] H. Lee, D. J. Mun, H. Lee, K. C. Lee, S. Hong, G. Kim, S. A. Lee, and S. Y. Chun, "Fidelity-preserving zero-shot diffusion models for highly ill-posed inverse problems in lensless imaging," *Image and Vision Computing*, vol. 165, p. 105786, 2026.